Solution Requirements

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction)
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*The system allows authorized users (researchers, policymakers, or administrators) to securely log in before accessing HDI prediction features.*	High
2	Authorization levels	*Different user roles (Admin, Researcher, User) have different permissions for viewing datasets, making predictions, and managing system data.*	High
3	External interfaces	*The system can import HDI datasets from CSV files or trusted public sources such as UNDP or the World Bank for analysis and prediction.*	Medium
4	Transactions processing	*The system accepts user inputs (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita), processes them through the machine learning*	High

		model, and returns the predicted HDI category.	
5	Reporting	*The system displays the predicted HDI category and generates reports or visualizations for analysis and decision-making.*	Medium
6	Business rules, etc.	*The prediction is based only on valid numerical inputs. The model classifies countries into one of four categories:* ***Very High, High, Medium, or Low HDI****.*	High
7	Compliance to laws or regulations	*The system follows data privacy and ethical AI practices and uses publicly available development datasets responsibly.*	Medium
8	*Other*	*The system provides an easy-to-use web interface for entering data and viewing prediction results.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	*The system should process user input and display the predicted HDI category quickly.*	*Prediction generated in* ***less than 2 seconds****.*

2	Scalability	*The system should support multiple users and future expansion with larger HDI datasets.*	*Supports **at least 500 concurrent users**.*
3	Security & Data Privacy	*Protect user information and prevent unauthorized access to the application.*	*Secure login, encrypted communication (HTTPS), and role-based access*
4	Reliability & Availability	*The application should remain available with minimal downtime.*	***99% system availability** during operation.*
5	Usability & Accessibility	*The web interface should be simple, responsive, and easy for users with different technical backgrounds.*	***Prediction completed in 3 simple steps with a user-friendly interface.***
6	*Other*	*The system should be easy to update with new datasets or improved machine learning models and should run on Windows, Linux, and macOS.*	Easy maintenance and cross-platform compatibility.